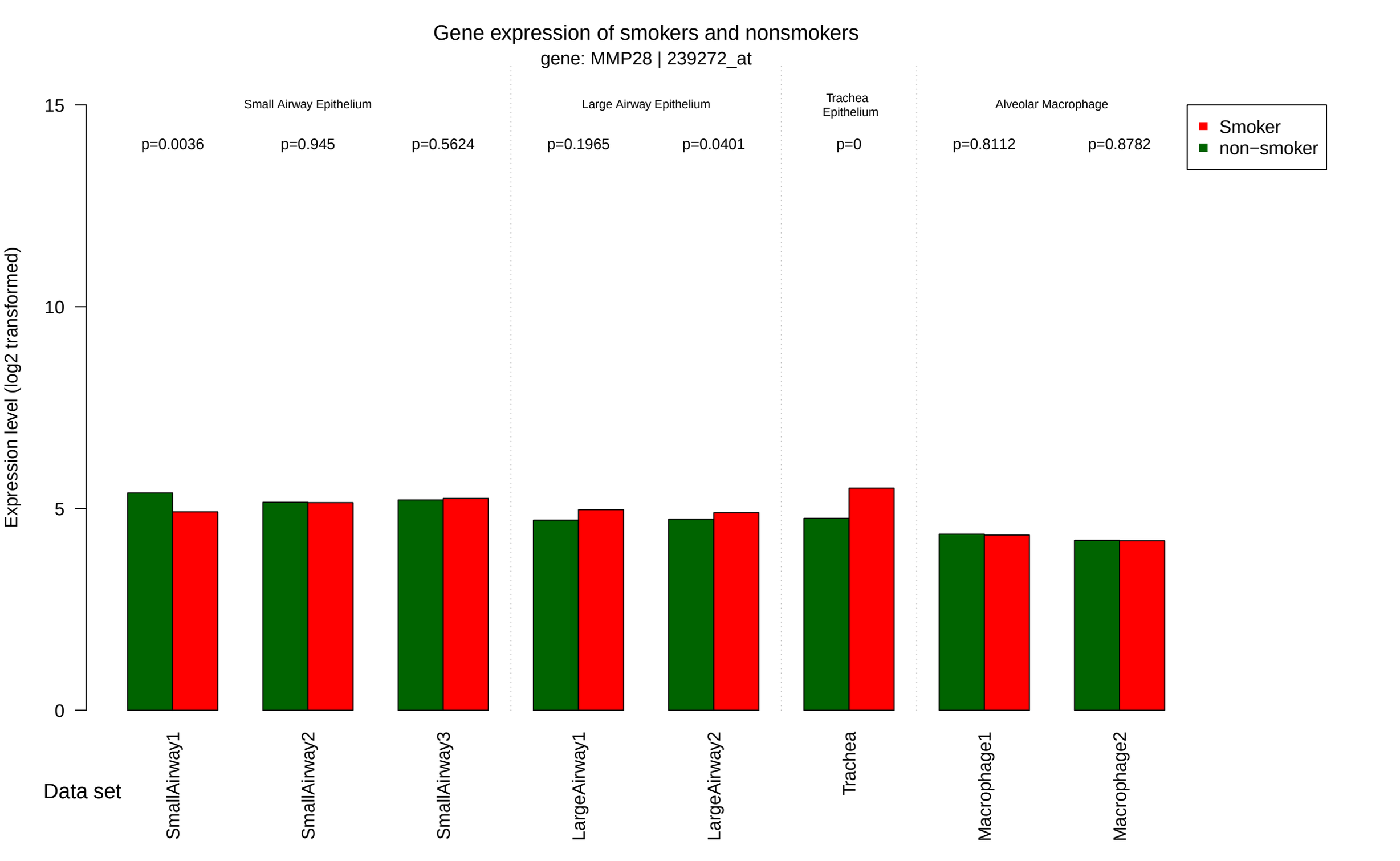
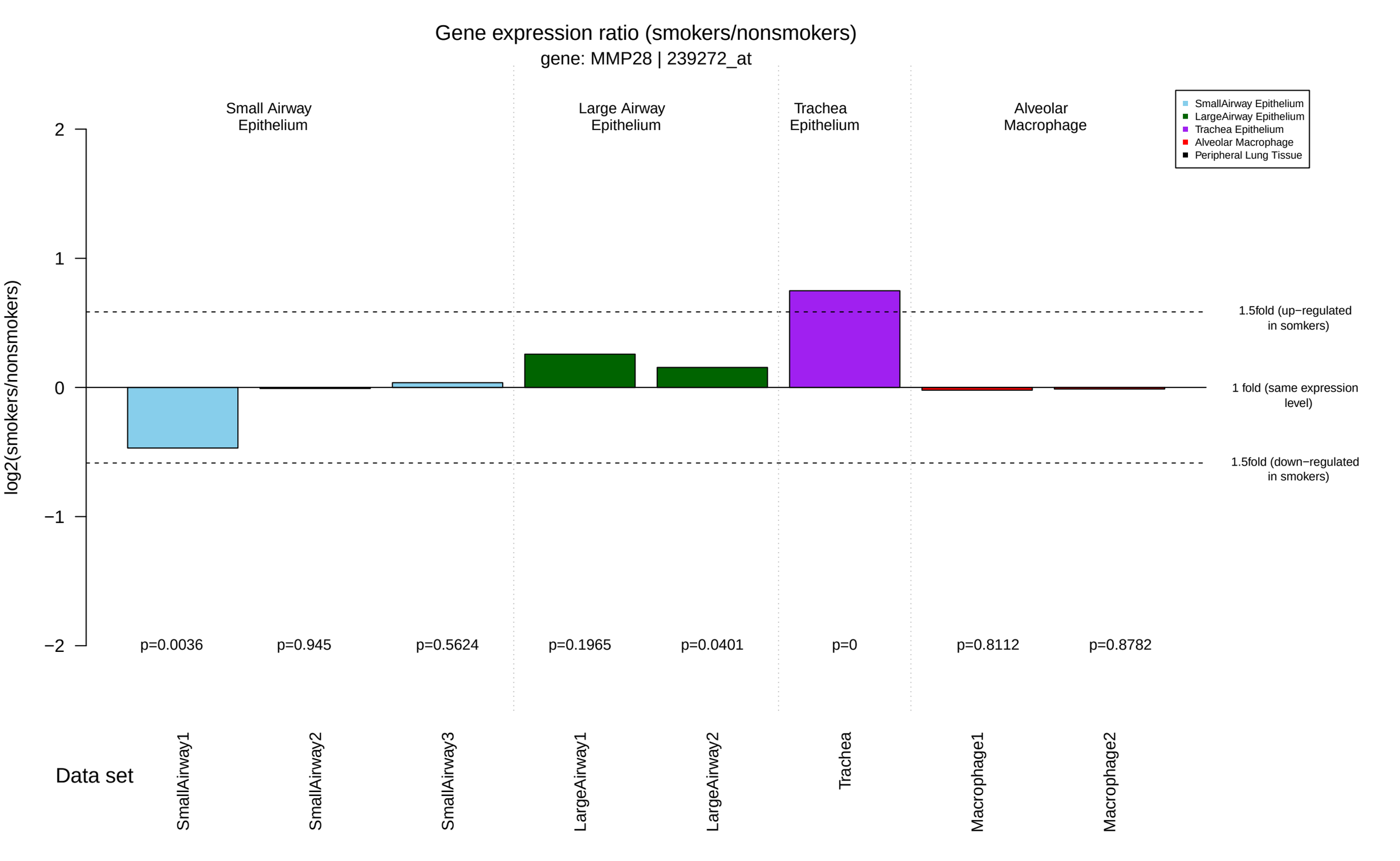
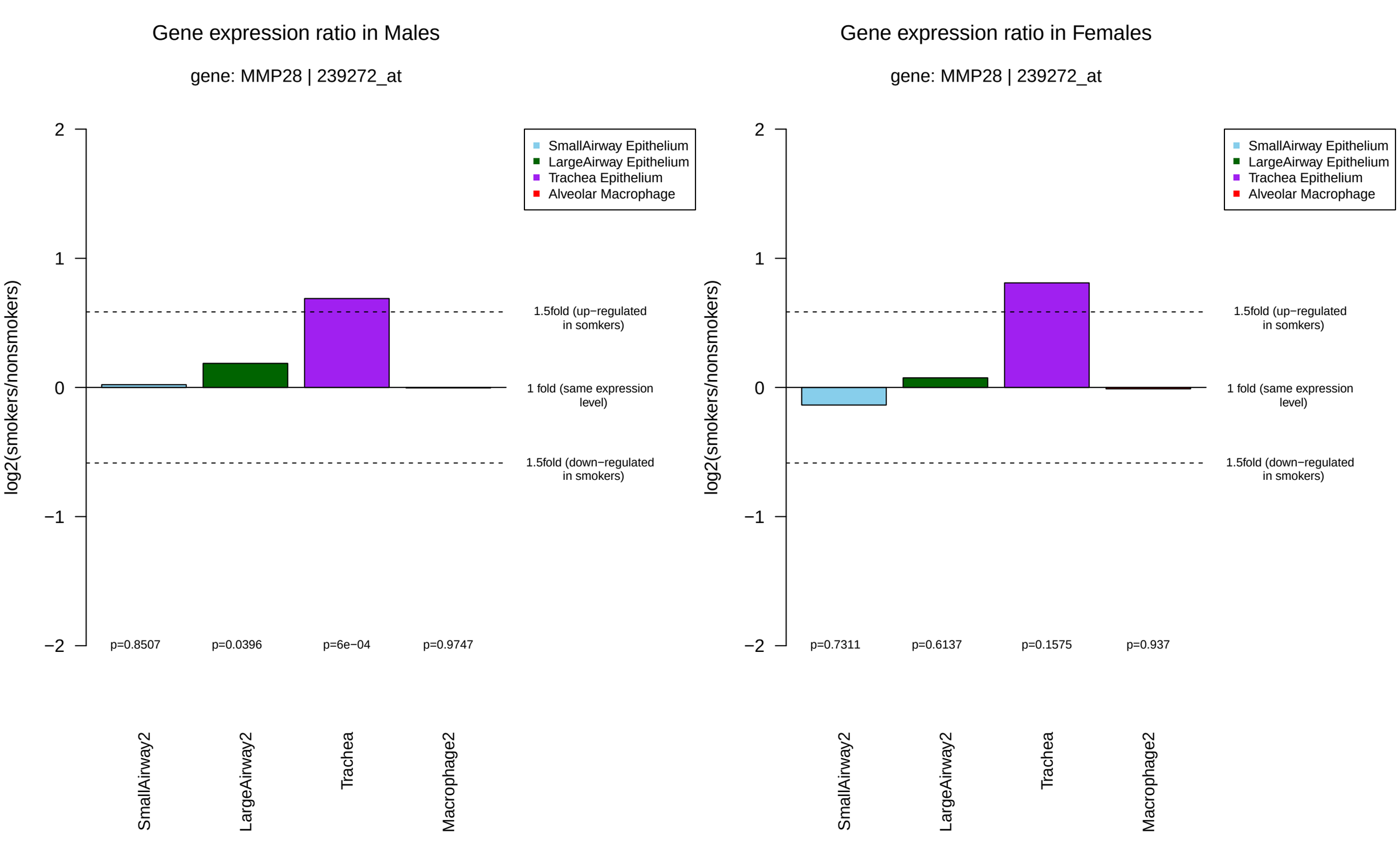
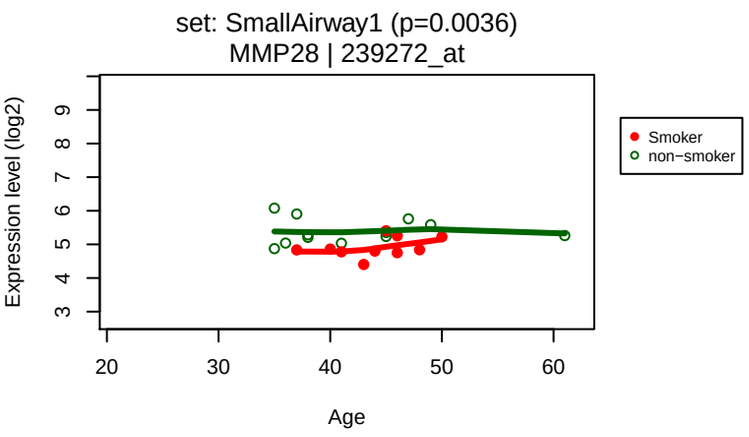
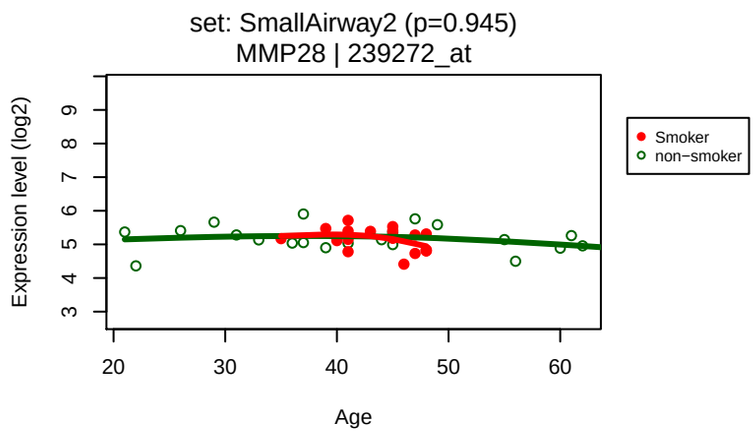
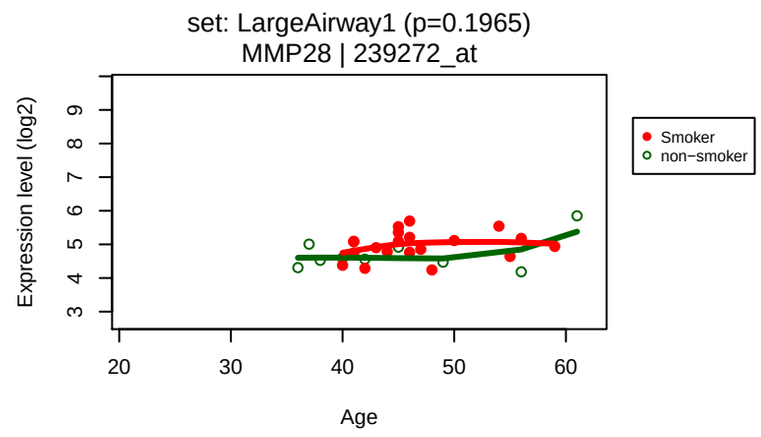

Gene expression of smokers and nonsmokers
gene: MMP28 | 239272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0036
p=0.945
p=0.5624
p=0.1965
p=0.0401
p=0
p=0.8112
p=0.8782
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

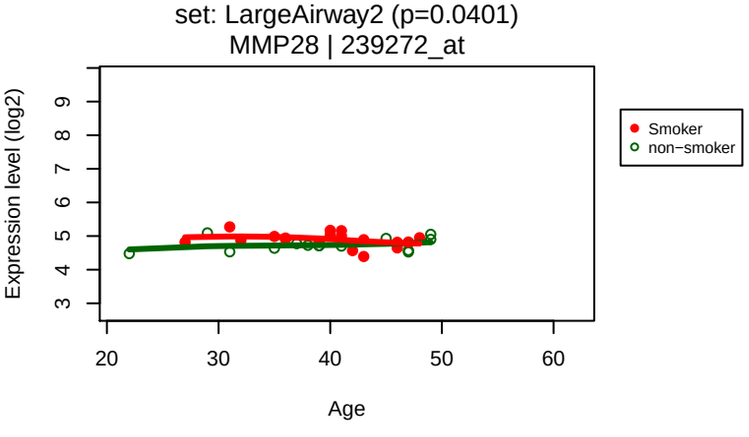

Gene expression ratio (smokers/nonsmokers)
gene: MMP28 | 239272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0036
p=0.945
p=0.5624
p=0.1965
p=0.0401
p=0
p=0.8112
p=0.8782
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

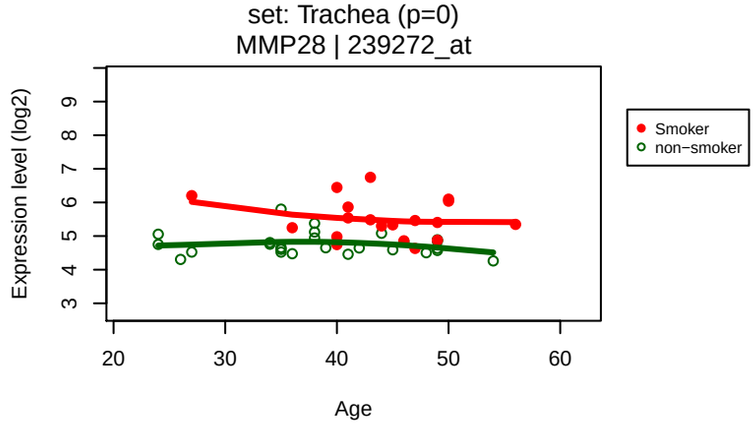

Gene expression ratio in Males
gene: MMP28 | 239272_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8507
p=0.0396
p=6e−04
p=0.9747
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP28 | 239272_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7311
p=0.6137
p=0.1575
p=0.937
SmallAirway2
LargeAirway2
Trachea
Macrophage2

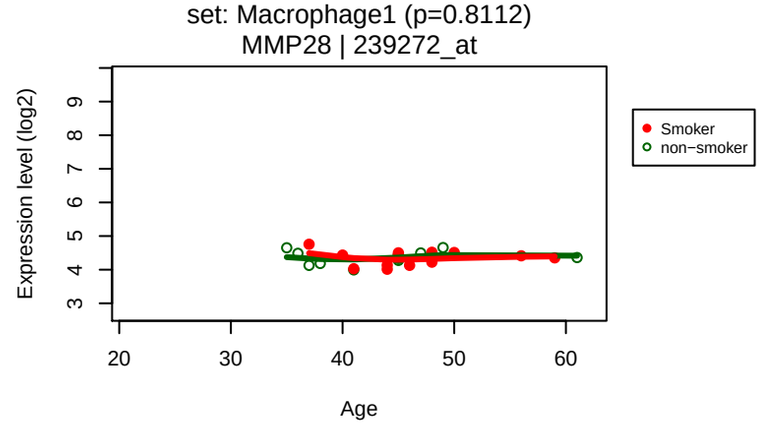

set: SmallAirway1 (p=0.0036)
MMP28 | 239272_at
Expression level (log2)
Age
Smoker
non−smoker

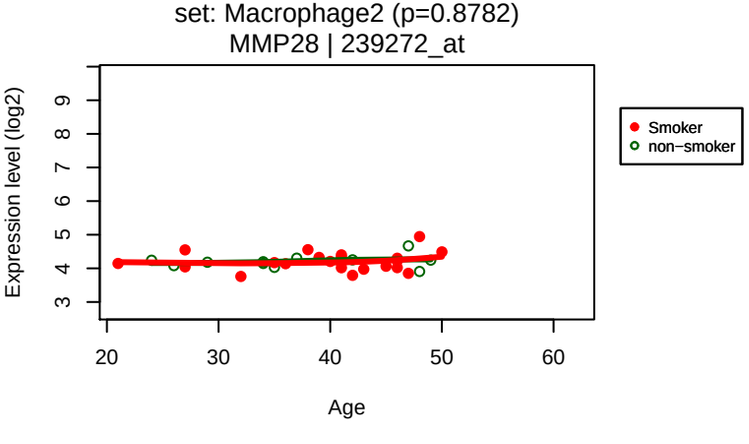

set: SmallAirway2 (p=0.945)
MMP28 | 239272_at
Expression level (log2)
Age
Smoker
non−smoker

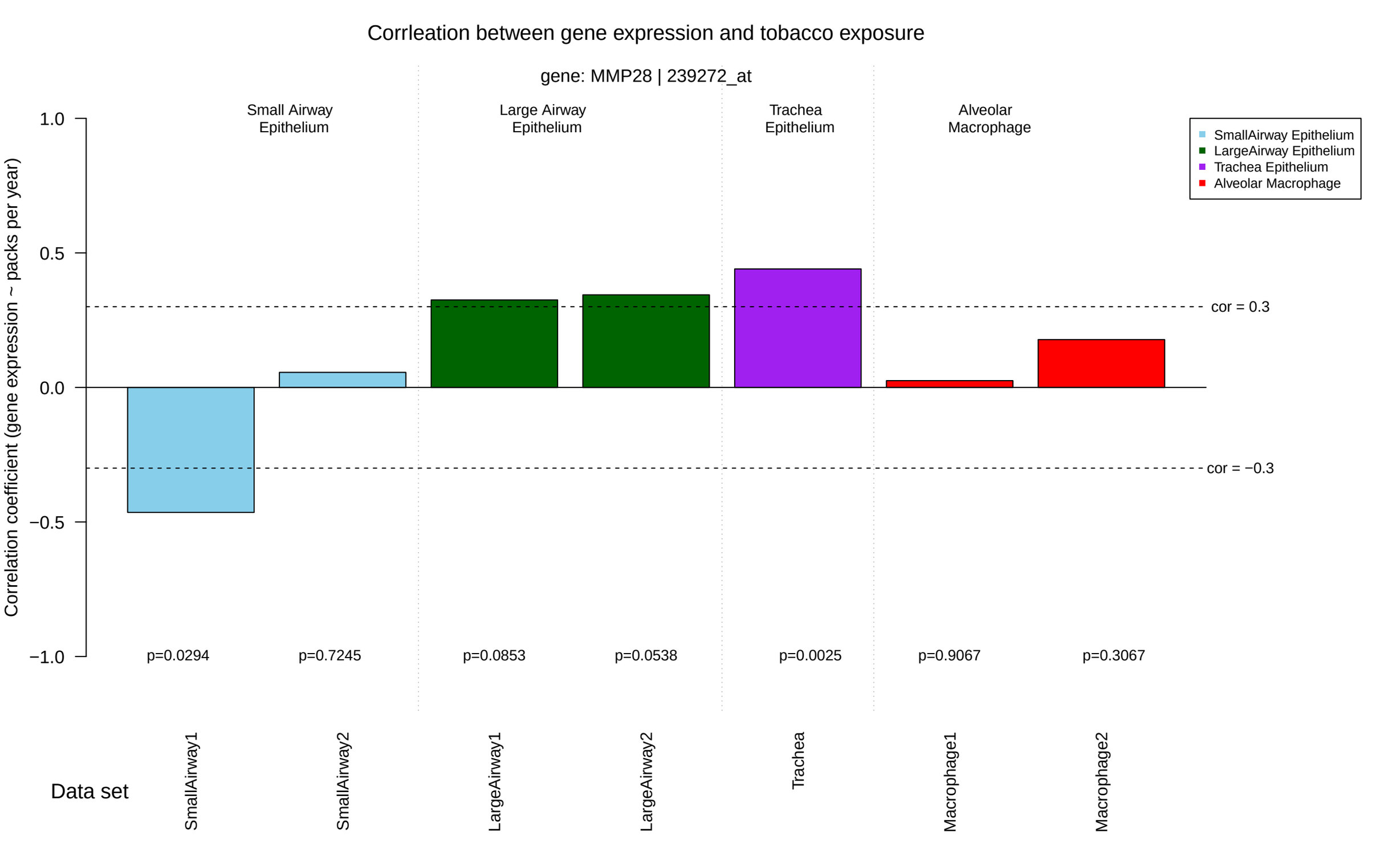

set: LargeAirway1 (p=0.1965)
MMP28 | 239272_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0401)
MMP28 | 239272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
MMP28 | 239272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8112)
MMP28 | 239272_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8782)
MMP28 | 239272_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP28 | 239272_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0294
p=0.7245
p=0.0853
p=0.0538
p=0.0025
p=0.9067
p=0.3067
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2